

MSc in Engineering – Data Science

Term Master-Thesis

Student	Roland Roccoaro
Advisor	Prof. Dr. Kenneth Ritley (ILV 100%)
Institute	BFH TI
Expert	Andreas Schlapbach, SBB –
Semester	AUT_2026 and SPR_2027
NDA (Non-Disclosure Agreement)	Yes
Type of work	MTh
ECTS	30
Subject	Enhancing User Confidence in Report Timeliness and Data Accuracy through Data Quality and Availability Checks

Abstract

This project builds on previous work on improving data quality and availability in the Swiss Post data warehouse environment. A successfully validated MVP for automated data quality checks will be **extended into a production-ready solution**. The project focuses on implementing automated validations for selected business-critical data and integrating their results into monitoring and reporting. The project is **conducted in close collaboration with the business users of the data and incorporates their feedback into the solution**.

Objectives

Mandatory

- Implement automated data quality and availability checks for the selected use case.
- Integrate **long-term monitoring data into appropriate visualisations**.

Required

- Integrate **user feedback into the solution**.

Optional

- Investigate additional data quality checks and use cases that could be integrated into the solution.
- Integrate the data into **additional automated monitoring streams**.

Literature

- [1] Specialisation Project 1 MTE7101 Enhancing User Confidence in Report Timeliness and Data Accuracy through Data Quality and Availability Checks, Roland Roccoaro, 11.01.2026
- [2] Specialisation Project 2 MTE7102 Automating data quality validations in Amazon Athena: An MVP based approach, Roland Roccoaro, 28.07.2026

Project registrations by email from 16 April (AUT) or 16 October (SPR)
Project proposal by email Deadlines: End of KW 40 (AUT) or KW 10 (SPR)

Project start	14.09.2026
Report delivery*	End of Spring Term 2027 KW31
Presentation date*	End of Spring Term 2027 KW33
Grade delivery*	End of Spring Term 2027 KW33 provisional mark when they give feedback.)

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Assessment	15' public presentation; 5-10' Q&A; project report; defence of the thesis ~30'
Deadlines	End of autumn term KW 3 (Report delivery) / KW 7 (Grade delivery) End of spring term KW 31 (Report delivery) / KW 33 (Grade delivery)